

1 Problem Description

p, q, r are random integer between 0, 6 inclusive.

these values represent the ratio of angle of a triangle inside a circle with its vertices on the circle's circumference.

what is the probability that the centre of the circle isn't in the triangle/on the triangle edge.

2 Problem Solution

$$k(p + q + r) = 180$$

The largest angle must be greater than 90 degrees following something like the circle theorem that states the angle is 90 degrees when it line passes through the centre of the circle.

This means that if p is the largest angle, then:

$$\begin{aligned} \frac{p}{p + q + r} &> 0.5 \\ \implies p &> (q + r) \end{aligned}$$

The dice can be modeled as a discrete random distribution. So the PGF would be:

$$\begin{aligned} G_p(t) &= G_q(t) = G_r(t) \\ &= \frac{1}{6}(t^6 + t^5 + t^4 + t^3 + t^2 + t) \end{aligned}$$

And then $q+r, p+q$ and $p+r$ can be modeled by the PGF:

$$\begin{aligned}
G_q(t) \times G_r(t) &= G_q(t) \times G_p(t) = G_p(t) \times G_r(t) \\
&= \left(\frac{1}{6}(t^6 + t^5 + t^4 + t^3 + t^2 + t) \right)^2 \\
&= \frac{1}{36} (t^6 + t^5 + t^4 + t^3 + t^2 + t)^2 \\
&= \frac{1}{36} (t^6 + t^5 + t^4 + t^3 + t^2 + t) (t^6 + t^5 + t^4 + t^3 + t^2 + t) \\
&= \frac{1}{36} (t^{12} + t^{11} + t^{10} + t^9 + t^8 + t^7) \\
&\quad + \frac{1}{36} (t^{11} + t^{10} + t^9 + t^8 + t^7 + t^6) \\
&\quad + \frac{1}{36} (t^{10} + t^9 + t^8 + t^7 + t^6 + t^5) \\
&\quad + \frac{1}{36} (t^9 + t^8 + t^7 + t^6 + t^5 + t^4) \\
&\quad + \frac{1}{36} (t^8 + t^7 + t^6 + t^5 + t^4 + t^3) \\
&\quad + \frac{1}{36} (t^7 + t^6 + t^5 + t^4 + t^3 + t^2) \\
&= \frac{1}{36} (t^{12} + 2t^{11} + 3t^{10} + 4t^9 + 5t^8 + 6t^7 \\
&\quad + 5t^6 + 4t^5 + 3t^4 + 2t^3 + t^2)
\end{aligned}$$

So now we just need to find the probability that the largest angle is greater than the sum of the other 2 angles. To do this I can just check the cases where one of the angles is greater than the other 2 (as they are all same distributions). So assuming p is the largest angle ratio, I

should have:

$$P(q + r = i) = \begin{cases} \frac{i-1}{36} & 1 \leq i \leq 6 \\ \frac{13-i}{36} & 7 \leq i \leq 12 \end{cases}$$

$$P(p > i) = \begin{cases} \frac{6-i}{6} & 1 \leq i \leq 6 \\ 0 & 7 \leq i \leq 12 \end{cases}$$

$$P(p > q + r) = \sum_{i=1}^{12} (P(q + r = i) \times P(p > i))$$

$$\begin{aligned} P(p > q + r) &= \sum_{i=1}^{12} (P(q + r = i) \times P(p > i)) \\ &= \sum_{i=1}^6 \left(\frac{i-1}{36} \times \frac{6-i}{6} \right) + \sum_{i=7}^{12} \left(\frac{13-i}{36} \times 0 \right) \\ &= \sum_{i=1}^6 \left(\frac{(i-1)(6-i)}{216} \right) \\ &= \sum_{i=1}^6 \left(\frac{-i^2 + 7i - 6}{216} \right) \end{aligned}$$

Now decompose and evaluate summations:

$$\begin{aligned}
&= \frac{-1}{216} \sum_{i=1}^6 (i^2) + \frac{7}{216} \sum_{i=1}^6 (i) + \frac{-6}{216} \sum_{i=1}^6 (1) \\
&= \frac{-1 \times 6}{216 \times 6} (6+1)(12+1) \\
&\quad + \frac{7 \times 6}{216 \times 2} (6+1) \\
&\quad + \frac{-6 \times (6-1)}{216} \\
&= \frac{13}{108}
\end{aligned}$$